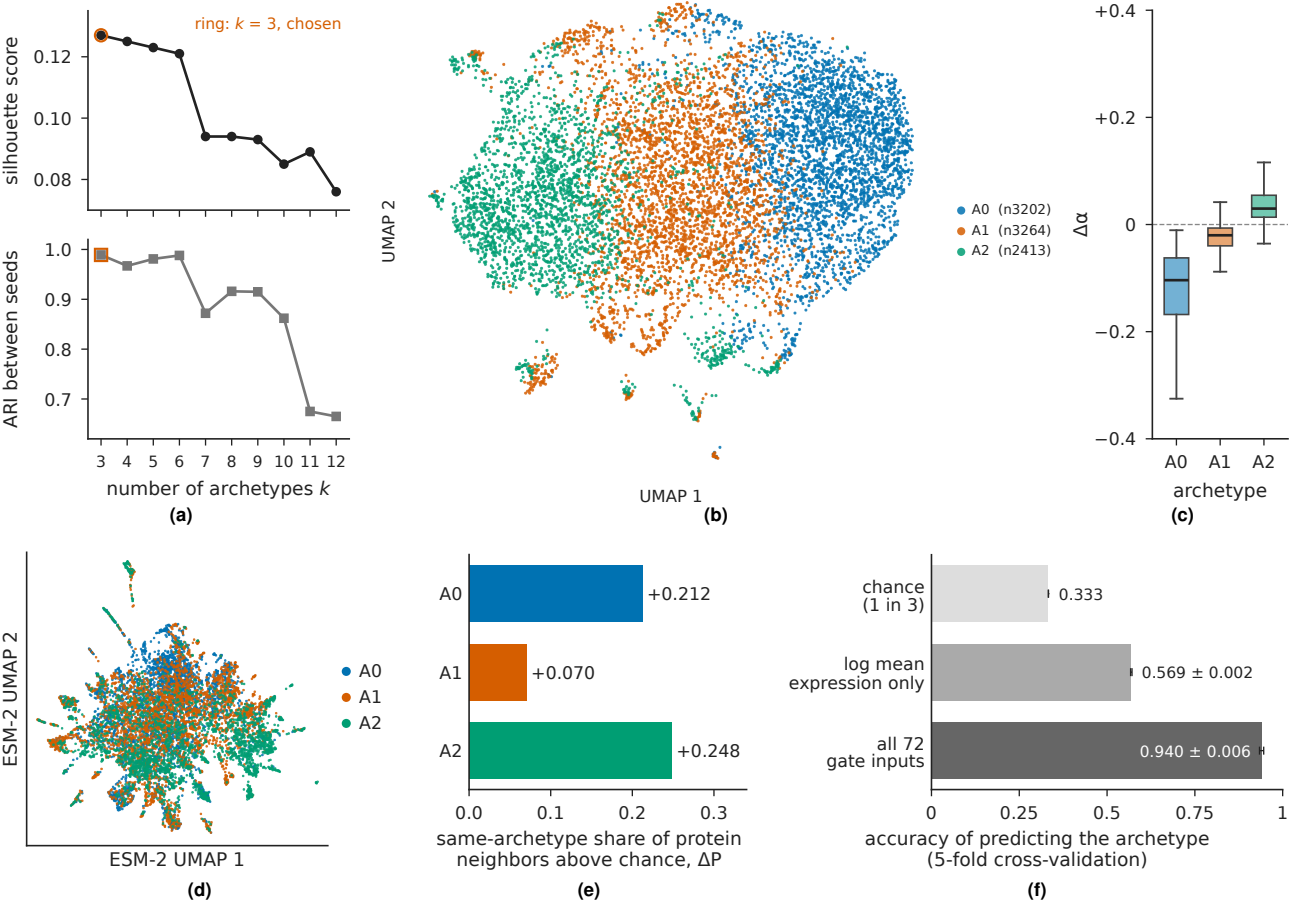

ring: k = 3, chosen
silhouette score
0.12
0.10
0.08
ARI between seeds
1.0
0.9
0.8
0.7
3 4 5 6 7 8 9 10 11 12
number of archetypes k
(a)
UMAP 2
UMAP 1
A0 (n3202)
A1 (n3264)
A2 (n2413)
(b)
+0.4
+0.2
0
−0.2
−0.4
Δα
A0 A1 A2
archetype
(c)
ESM-2 UMAP 2
ESM-2 UMAP 1
A0
A1
A2
(d)
A0
A1
A2
+0.212
+0.070
+0.248
0.0 0.1 0.2 0.3
same-archetype share of protein neighbors above chance, ΔP
(e)
chance (1 in 3)
0.333
log mean expression only
0.569 ± 0.002
all 72 gate inputs
0.940 ± 0.006
0 0.25 0.5 0.75 1
accuracy of predicting the archetype (5-fold cross-validation)
(f)